



Eye Tracking Glasses

Pocket Guide



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Introduction

From the very beginning of the technology, eye tracking glasses have been promoted as the eye tracking solution for the real world. The glasses promise to bring research out of the lab and into naturalistic settings. While that was true to a point, eye tracking glasses have not always been the confinement-free research tool that was promised, and more often than not research in the wild was still quite restricted in terms of mobility and accuracy.

Today it is feasible both technically and economically to bring eye tracking out of the lab to conduct studies in the real world. We are now seeing the usability that was originally envisioned begin to emerge. Higher accuracy, increased robustness of the design, and overall higher design flexibility are paving the way for scientific advancement in several fields that have up until recently only been able to apply eye tracking glasses in a limited scope.

In this guide, we will explain the what, how, and why of eye tracking glasses - explaining everything you need to know to get started with your own experiments.

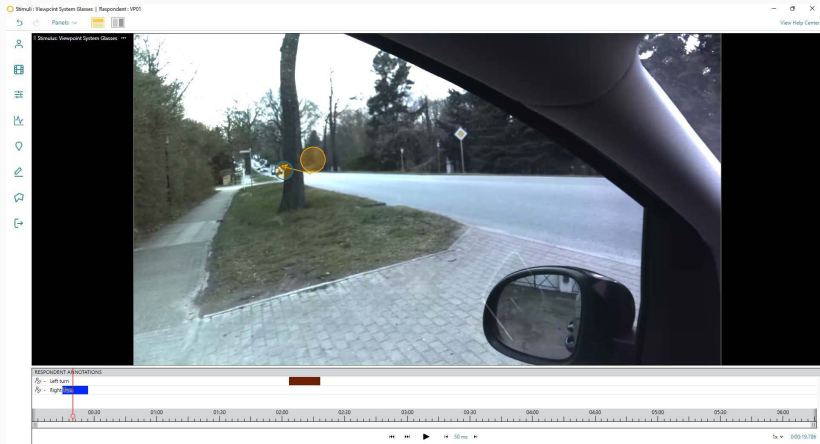


Eye tracking glasses are usually referred to as such, but they are also sometimes known as head-mounted eye tracking or mobile eye tracking.

In striving to move data collection from the lab and into the wild, eye tracking technology companies have continued to evolve their devices to be more accurate and robust. The increased mobility of eye tracking solutions means that subjects no longer have to be tethered to a screen which enables higher ecological validity in research designs. Eye tracking glasses, where the eye tracking technology is built into a pair of glasses and worn by the user instead of mounting to a screen, gives researchers the opportunity to collect data such as attention, gaze patterns, and saliency to gain insights into visual perception, and cognitive performance, across a variety of research domains.

In the not-so-distant past, eye tracking glasses were bulky devices, braced to the head and far from natural to wear. Today, after 20 years of innovation, enabled by faster computers, smart phones, camera miniaturization, and machine learning - eye tracking glasses have acquired a more natural appearance, very close to the form factor of a normal pair of eyeglasses. The improvement of the technology not only makes it smaller and cheaper, it also allows for taking the technology from inside the controlled environment in a lab to the outside world, to be used for research in a natural environment.





An example of eye tracking glasses applied in automotive research. The driver is about to make a right turn onto a busy road; is she noticing the oncoming traffic? A researcher has annotated the recording timeline to indicate important events such as left and right turns (screenshot from iMotions).



What can eye tracking glasses be used for?

As a consequence of the improved form factor and reduced costs, commercial and academic scientists are increasingly using eye tracking glasses as a part of their research. A wide range of application areas benefit from eye tracking glasses, including:

- » Psychology - dyadic interactions, autism, ADHD, Schizophrenia, Alzheimer's, Parkinson's.
- » Business - interviews and negotiations, presentation optimization
- » Medical - visual attention of doctors and nurses during tasks, X-ray image interpretation, diagnostics, simulation and training.
- » Marketing - In-store behavior, menus, billboards, product, logos
- » Architecture & Construction - visual attention in buildings, safety, urban environments
- » Computer Science - human computer interaction, gaze contingency, gaze control.
- » Transportation Simulation - visual attention across all modes of transportation: automotive, rail, aviation, etc. Driving performance, human computer interaction, safety, attention, distraction.
- » Human factors - general assessment and interventions of human performance, for example in sports, law enforcement and military

This guide will primarily focus on research using eye tracking glasses, but the hardware is also commonly used as a method to improve human performance (such as with human factors research). For the most part, the same considerations are necessary for both fields, so this guide should therefore be useful for anyone using eye tracking glasses either for research purposes or to improve human performance.

One of the really powerful aspects of using eye tracking glasses is its ability to easily be combined with other sensors, thereby further increasing our ability to assess and detect human states in ecologically valid environments. Such a multimodal approach can include using sweat response (electrodermal activity, EDA) and heart rate (electrocardiography, ECG) data to gain insights into how emotionally or physiologically arousing certain visual targets are in testing areas. This can include usage in stores, medical environments, or in training simulators. Additionally, metrics like EEG can be combined with glasses to understand complex cognitive states such as workload or drowsiness on a brain-based level. iMotions specializes in helping researchers move beyond single datastreams and into multi-sensor applications, in order to help researchers understand human behavior better.

One of the major challenges with using eye tracking glasses as a part of your

research is that the technology can be quite complex to use, and the data often requires many hours of analysis. However, these challenges can be circumvented by choosing the best hardware and software platform for your needs.

Measuring visual attention in natural environments

Scientists have studied eye movements for over 150 years, and in that span of time we have come to learn a lot about how our eyes work! In the following we will dive into some of the key concepts that are important for understanding eye tracking data.

Fixations and Saccades: Categorizing the behavior of our eyes based on how much they move

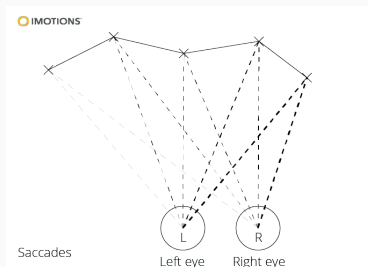
Despite over a century of research, scientists still debate one of the most fundamental concepts of eye movements (Hessels et al., 2018). In this guide, we will not discuss the details of this debate, but we encourage all current and future eye tracking researchers to partake in the discussion. It is also important to note that it is heavily debated whether it is appropriate to perform fixation analysis on eye tracking glasses data. The main reason is that while your eyes move, your head may also move. Because both the eyes and the head could be moving, you do not actually know if people are fixating on an object or if they keep looking at it while moving their heads. Owing to the debates and discussions in the field, we welcome researchers to avoid using black boxes for analysis, and report everything they can about their analysis, from pre-processing to the metrics used and of course the resultant statistical analysis.



Two of the main characteristics of eye movement behaviors are as follows ([Mahanama et al., 2022](#)):

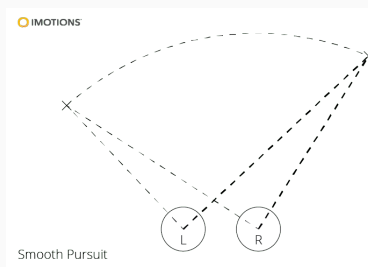
1. **Fixations:** fixations refer to when the eye is focused on a particular static object. In this period, the eyes move minimally (although they do have some tremor). During fixations, our fovea is focused on the object of interest, and this is where our primary visual processing occurs. Typically, what our fovea fixates on is what we visually perceive at a conscious level. Fixations can be calculated in different ways, and discuss this further (on page 33) when we talk about fixation filters.
2. **Saccades:** saccades refer to the process of the eyes moving rapidly from one fixation to another. In other words, saccades drive our visual attention from one object to the next, and, in contrast to fixations, we do not (consciously) perceive our visual environment during these saccadic eye movements. Saccades are repressed when we focus our visual attention onto stationary objects.

In sum, fixations and saccades are important for visual attention processing. Deficits in maintaining fixations (and repressing saccades) can be used as biomarkers for several clinical disorders.



Smooth pursuit: when our eyes track objects in the environment

Sometimes we fixate on a moving object, for example a ball on a field. In that process we are fixated on a moving object, leading to slower eye-movements called smooth-pursuit. Smooth pursuits are basically a series of short and slow saccades, but they have a lower latency (around 100 - 125 ms) compared to regular saccades (200-250 ms), and they are specifically referred to as smooth pursuit movements.



Vergence, divergence and estimating distance

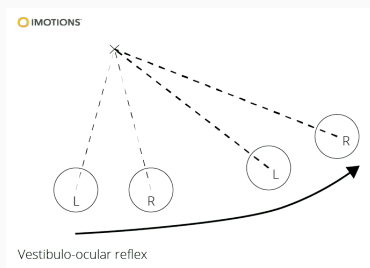
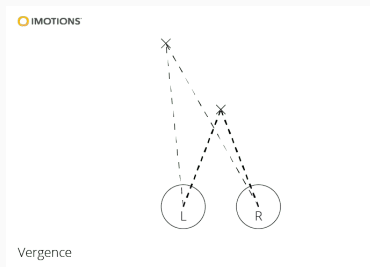
Most of the time, we perceive the world through binocular vision (meaning that both of our eyes are open and visually active). While we may not notice it, our eyes will sometimes move in opposite directions and to different degrees in order to focus on an object in space. For example, if an object is placed to the left of us, we can move our eyes (instead of our head) to focus on it. In doing so, our left eye will move fewer degrees than the right eye, and in total, the foveas from each eye are now closer together. This is called convergence, and typically occurs when an object is close by.

In contrast, when the object is farther away (but still off to the left), our left and right fovea will start moving farther away from each other. This is called divergence. Vergence is also a part of the [accommodation effect](#) where our eyes automatically focus on a nearby and subsequent far-off object.

Vestibular-ocular movements

Vestibulo-ocular movements are what allows us to see clearly even when moving our heads. When we walk, for example, our head moves around a great deal and the vestibulo-ocular movements compensate for this movement, keeping our [eyes on the intended visual target](#).

If this process didn't occur, we'd only be able to move our gaze direction by moving our head (much like the owl, whose eyes are essentially immobile



(Schwab, 2003).

Interestingly, this process continues to work even in the dark – showing that this process is controlled not by the perception of light but solely by the vestibular system.

The parallax eye: monocular vs binocular vision

If you close one of your eyes, and only perceive the visual world through one of them, it is referred to as monocular vision. As exemplified with the vergence effects, our eyes do not always move synchronously, which means that each eye perceives the depth of an object slightly differently, however, this becomes most apparent when we compare our monocular perceptions from the left and right eye. Therefore, your eye tracking

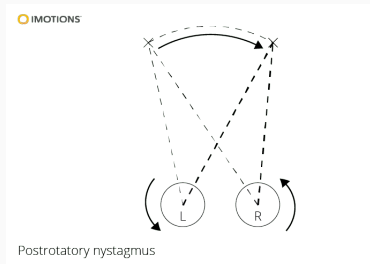
data will look different depending on whether you are tracking only the left or the right eye (monocular vision) or averaging the gaze point from both eyes (binocular vision).

Optokinetic response movements/ postrotatory nystagmus

Despite their intimidating names, these two forms of eye movements aren't so complicated. The optokinetic response and postrotatory nystagmus are in essence reflexive combinations of saccadic and smooth pursuit movements.

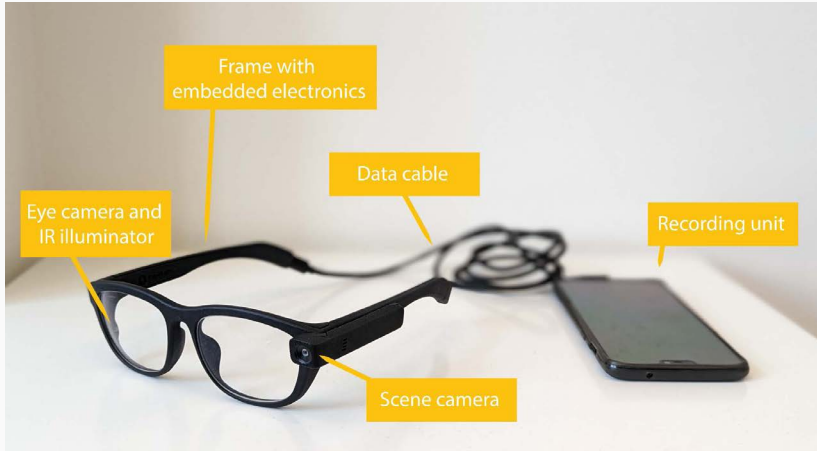
The optokinetic response is a small movement of the eyes that track a stimulus. This often occurs when looking at a moving object that then moves out of the visual field. Think of looking out of the window when in a car – it's possible to track the objects, but as they move out of the visual field, your eyes can make a corrective, horizontal jump to return to their original position (Mustari and Ono, 2009).

Postrotatory nystagmus is what happens if you spin around in one spot too much. The eyes, compensating for the rapid movement, begin to move in the opposite direction to travel. This is also a phenomena that occurs in the absence of light, showing the involvement of the vestibular system in initiating this movement. Both of these eye movements are forms of nystagmus, although not pathological (they are present in normal eye movement behavior).



Final thoughts

While it is scientifically interesting to understand the oculomotor principles underlying different types of eye movements, the information is typically not relevant for most research questions in the fields of communication, business, marketing and many areas of psychology and neuroscience. When we want to understand visual attention (and the lack of it), fixation and saccade metrics are typically sufficient. Let's now turn to how eye tracking glasses work to measure our eye gaze.

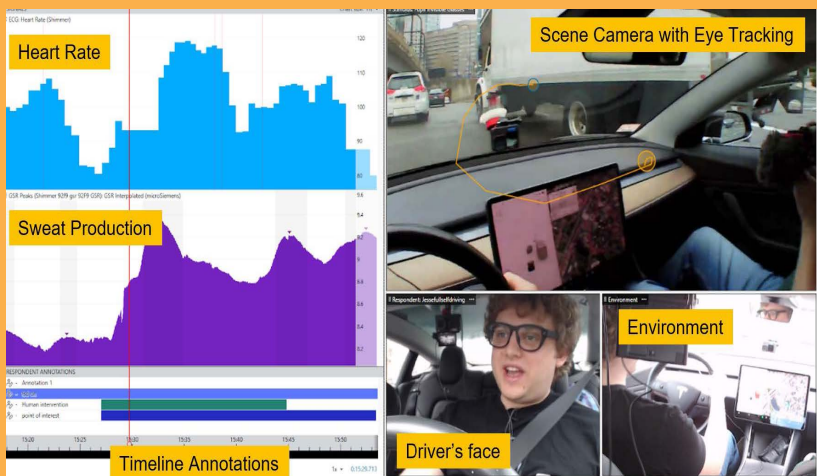


Hardware components

Eye tracking glasses hardware typically includes the following elements:

- A frame to host the sensors and control electronics.
- Eye cameras that are embedded in the rim of the frame, pointing towards the eyes in order to capture the eye movements. The cameras are typically sensitive in the near infrared spectrum.
- Lights that illuminate the eyes in the frequency range of which the eye cameras are sensitive (typically infrared).
- A front-facing camera (often referred to as a scene camera). This camera captures the environment.
- Recording unit. Typically connected to the glasses frame using a cable such as USB. The recording unit records and processes data from the cameras. It can perform the transformation from video to gaze data, as well as provide a live view of eye tracking data superimposed on the scene camera view.
- The recording unit may also offer a way to wirelessly transmit data to a client so that video and gaze data can be observed live, remotely.

- Customization options to optimize the fit to the respondent and environment:
 - » Nose pieces: In order to ensure optimal positioning of the eye cameras the glasses can have a range of nose pieces to ensure comfort and fit.
 - » Prescription lenses: It can be difficult to put eye tracking glasses on top of existing glasses a participant may have (the reflection from glasses can in some cases also interfere with data quality). As a result, the eye tracking glasses may offer a range of lenses that can be inserted to match the participant's eyesight.
 - » Sunglasses: To minimize the effect of intense sunlight on the participant and allow for the cameras to pick up the eye movements more effectively.



An example of how eye tracking glasses can be used in a real world scenario in combination with multiple modalities (physiology and video feeds) to understand human behavior, cognition, emotion, and performance. A driver is monitored using eye tracking glasses while driving. Heart rate and sweat gland activity is measured with ECG (electrocardiography) and EDA (electrodermal activity) devices respectively. This allows for identification of the moments that are most stressful when interacting with the autonomous vehicle.

Credit: Now You Know Youtube channel; Screenshot from iMotions' software.

How do eye tracking glasses work?

All eye tracking glasses aim to assess where the gaze is in relation to the scene camera - which is the front facing camera that captures what the person's head is facing towards. The eye tracking data measured by the glasses is mapped onto the video captured by this scene camera, just like a screen-based eye tracker maps the data onto the computer screen.

As mentioned above, the eye tracker itself is embedded into the frame of the glasses. There are two main types of eye trackers. Some rely entirely on image processing to detect *the iris and the pupil*, others use a reflection of infrared light on the *cornea to identify the pupil* and track where it moves. While the former still uses infrared light to illuminate the pupil, and the latter method still uses image processing, the key difference is whether they use corneal reflections to estimate gaze or not. Until recently, eye trackers based on corneal reflections were thought to be superior and more accurate. However, the gap between these two types is rapidly closing with faster and newer advances in machine learning.

Which eye tracking glasses should I choose?

If you plan to conduct research using eye tracking glasses, it is good to think about what requirements you have for the performance of the hardware.

Here are some features of the system you may want to consider:

Gaze data sampling frequency.

Does your research question require a certain data rate for gaze data to be answered? The sampling rate typically varies across hardware products from 60 to 200 Hz.

Scene camera.

What is the resolution of the scene camera? Most hardware platforms support HD equivalent resolution and update frequency of 30 Frames Per Second (FPS). Most systems also support recording of sound.

Also note what options are available to adjust the focus of the lens to emphasize objects that are either near or far. Additionally you may also like to know how the system handles automatic adjustment of gain due to changes in light conditions in a natural environment. Furthermore Field of View (FOV) is essential - FOV is usually measured in degrees of visual field that the camera is able to "see".

Battery lifetime and management.

How long a recording can you do on one full charge? Can you swap batteries easily to avoid having to wait for charging between respondents? Can the device operate without a battery plugged into an external power source?

Accuracy.

Does your research depend on the device performing above a certain, well defined threshold for accuracy? Note that in order to achieve greater accuracy, the device may require a manual calibration. With recent technological advancements, manual calibration has become seamless and easy to implement while also increasing accuracy.

Population.

If your participant population is adult most systems will work fine as they are typically designed for this group. If you test on infants you want to make sure the form factor of the glasses will fit a smaller head size and that its calibration is specialized for infant eyes.

Signals apart from gaze data, what signals does it provide? Some signals that you may be interested in are pupil size, eyelid position, blink detection, IMU (accelerometer, gyroscope).

Multimodal recordings.

Does the eye tracking glasses platform have a way to be combined with data from other sensors such as skin conductance (EDA/GSR), Heart Rate (ECG, or PPG), respiration, brain activity (EEG), voice, additional video

feeds and so on.

Robustness.

What kind of environment will the participant be in? Do you expect the equipment to be subjected to a rough movements, shaking, dirt, impacts of various kinds? For example if you are doing research in sports performance, such as basketball or soccer, you should look for hardware that has a rugged design.

Slip compensation.

If the glasses slip down the nose over time, is the system able to compensate for the change in camera-eye configuration?

Cost.

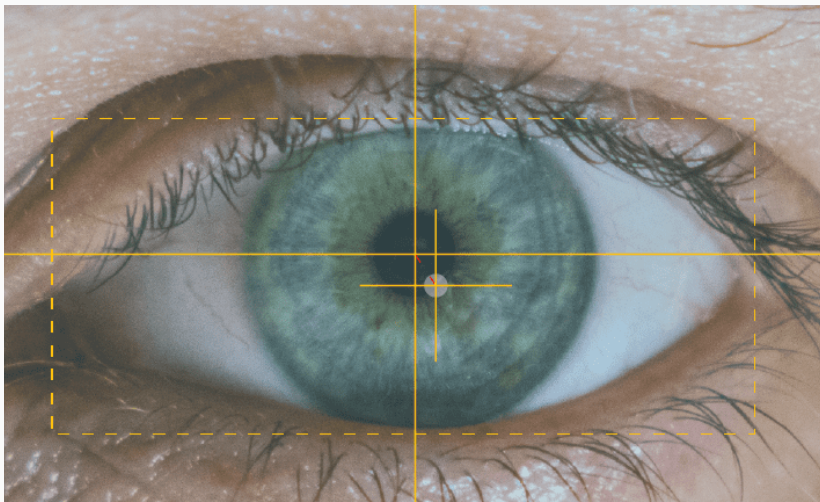
How much does the hardware cost? Software cost will also have to be considered. Your exact budget and requirements will ultimately determine which system is right for you.



Some of the vendors in the market

- | | |
|-----------------------|-----------------------------|
| • Pupil Labs | Neon, Invisible, Pupil Core |
| • ViewpointSystem | VPS19 |
| • Argus Science | ET Vision |
| • Adhawk Microsystems | Mindlink |
| • Ergoneers | Dikablis |
| • Positive Science | UltraFlex, METL |
| • Kexxu | Openeye |
| • ISCAN | ETL100H |
| • Pertech Solutions | EyeTechSensor |
| • 7invensun | aSee |
| • Tobii | TG2, TG3 |
| • SeeTrueTechnologies | |

In addition to “regular” eye tracking glasses, companies are also starting to develop eye tracking glasses with augmented reality, for example the Varjo XR-3, HoloLens, and MagicLeap. Although the technology is still advancing, the use of augmented reality is expected to increase exponentially as Apple, Google, Facebook and other big players continue to improve the technology. Pairing augmented reality with eye tracking capabilities gives researchers the capability of tracking visual attention in both real and augmented environments in a way that is more ecologically valid than doing research with virtual reality headsets or simulators. This further underscores the importance that eye tracking technology will continue to play in research.



Software components

With the core tools at hand, you're perfectly equipped to track the basics. You can now find out where, when and what people look at, and even what they fail to see. So far, so good.

Now how about pushing your insights a bit further and stepping beyond the basics of eye tracking?

Choosing the right hardware is essential for getting the right data for your research question, it is equally important to consider the software you will use for data collection and analysis. In this section we will go over the main capabilities you need to consider.

But before we move on, it is first important to distinguish between the software you will receive with the hardware and the software you will use for data collection and analysis. Most eye tracking glasses will come with dedicated software. This software will usually allow you to modify hardware settings, visualize the live recording and, start/end and store recordings. In most cases, hardware-specific software doesn't offer analytical features, advanced study design options, or visualizations. For the most part, the software is not built for research, but simply to control the hardware. In the following section we will be focusing on what to consider when it comes to the software you use for data collection and analysis. For some tasks, the eye tracking glasses specific software might be sufficient, while others will require a separate software.

Controlling the recording

It sounds easy to start and end a recording, but with eye tracking glasses that is not always the case. As mentioned above, most eye tracking glasses come with their own software from which you can control the recording. If you use this option, then you need to get an analysis software that can export existing recordings in the appropriate file format. However, there are multiple downsides to this option. First, you are, depending on the glasses, adding an unnecessary step of first collecting the data in one software

and then subsequently importing it into a subsequent analysis software. By collecting and analyzing your data in one software you minimize your workload. Secondly, if you intend to use additional biosensors for your eye tracking glasses research you will have more challenges synchronizing the data. If you collect all of your biosensor data in the same software, then you circumvent the issue of trying to align the different data streams.

For the recording, you also want to consider the experimental design and the choice of hardware. Will you have

wired setup or will the eye tracking glasses connect to the software via WiFi? If you are using eye tracking glasses that require an external unit for controlling the recording, how does that influence your participants and data collection validity? If you are using a wired set of eye tracking glasses, is it possible to have a PC operated by the researcher close to the participant during data collection? This goes to show that you need to consider what your experimental design involves before choosing the hardware and the data collection software.

The calibration

It is also important to consider how the calibration is done and, importantly, visualized. You should always verify the calibration quality by visually inspecting the gaze data. This can be helped by asking the participant to look at a known target in the environment. Live viewing of the calibration will help you tremendously in troubleshooting calibration issues, and speeds up the experimental preparation process. Some eye tracking glasses use machine learning algorithms for calibration, which means that you do not have to consider these points although we highly recommend that you always have the capability of inspecting the eye tracking data quality and/or use an alternative manual calibration.

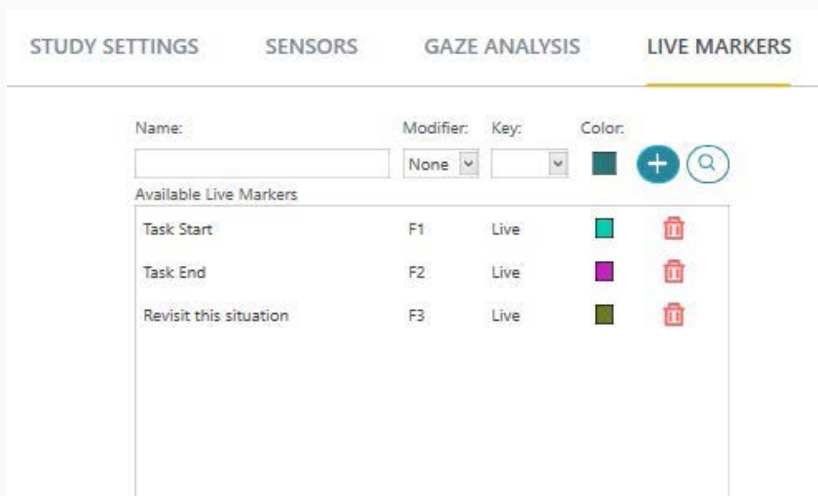
Live view of the recording

Having the option for live view during the actual recording is essential. Most systems support remote live view of the recording in order to monitor data collection and allow for qualitative

observations during the experiment. However, it is important to note that the broadcasted live view is not the data that will be used for analysis as it can suffer from latency issues and lower grade video quality due to network streaming bottlenecks. Instead, we want to analyze the recording that is ultimately saved onto the recording unit as this will have the highest quality. This locally saved recording of the raw video and gaze data is then transferred later to PC for analysis after the experiment ends either over the network from the recording unit to the PC or by inserting a SD card (memory device) from the eye tracking glasses recording unit into the PC. If you go with eye tracking glasses from Pupil Labs, then you have the third option of uploading from the recording unit (a cell phone) to a cloud service where the data can be reviewed and downloaded to your analysis solution.

Behavioral coding during the recording

During the recording, it can be beneficial to add live markers into the data. The live markers encode various predefined events, e.g. task start and task stop. When reviewing and analyzing the data after the fact, these live markers can speed up the research as one can more easily find the essential moments in the recording. This capability is often not possible in the eye tracking glasses dedicated software, and requires a data collection software that also enables subsequent analysis. Below is an example of how live markers can be used and implemented in iMotions.



In iMotions software you can set up live markers that encode for certain types of events during the recording.

Synchronize with other data sources

As mentioned earlier, your choice of data collection software will determine how easy it is for you to synchronize your eye tracking glasses data with other biosensors. There are two general methods to take, and which one fits you best will depend on your research question and design.

1. Synchronize the different biosensor streams after data collection

If you choose to collect the eye tracking glasses data onto the hardware specific recording unit, then you can align the data streams from other sensors after the data collection

is over. In this case, you need some kind of shared stimuli that can mark exactly when the recording started for both devices. Traditionally, this has been done with a TTL pulse, but newer and easier options now exist. For example, in iMotions you can use a “synchronization clock” where you can seamlessly time stamp the beginning of the data collected from both the eye tracking and additional biosensors. In this case, you would collect all the other datastreams (e.g. GSR/EDA and HR) directly into iMotions while the eye tracking recording would be done on the hardware specific recording unit.

2. Synchronize the different biosensor or data streams during the recording

Eye tracking glasses data is valuable on its own, but adding additional data

streams can amplify the insights and data quality. Examples of valuable datastreams to add to your eye tracking glasses experiment include additional video feeds, live behavioral coding, and, of course, biosensors. The additional video feeds are not just helpful but also necessary for understanding how a participant moved around during an experiment, and live behavioral coding allows you to record the behaviors of interest during the data collection. This saves you time and improves the accuracy.

There are several biosensors that are regularly combined with eye tracking glasses, including GSR and ECG (see more examples below). Adding these additional sensors gives you an ability to understand how an experience, that the participant paid visual attention to, affected their physiology. Did the experience make them more aroused? Did they get a lower heart rate variability? These insights provide a valuable layer to your eye tracking data analysis.

In all these scenarios (video feeds, live behavioral coding and additional biosensors), you need to consider whether your software can collect data from multiple biosensors at the same time, import multiple video feeds and whether it can support live behavioral coding. Moreover, you need to consider the usability of said software and whether the subsequent analytical tools are appropriate for your ultimate research questions (see below what to consider in the analysis of eye tracking glasses data).

How to collect good data

There are specific tools you will benefit from having access to. These include: annotations, areas of interests, heatmaps, fixation filters, and key eye tracking metrics including dwell time, revisit count and time to first fixation. We will go into each of these tools below. This section will give you guidelines for how to collect good quality data in order to maximize the data yield and avoid unpleasant surprises once you get to the analysis phase of the research. Before starting this section, it's important to keep in mind that your choice of hardware and research question will influence your optimal data collection strategy. Moreover, we recommend that you have a plan for how to get access to participants.

The order and content of the following steps can vary depending on what your exact research setup looks like.

Pilot Test

Before inviting any participants to your study, always pilot test your protocol using yourself or a colleague as participants. Pilot testing involves everything from setting up the equipment to analyzing the data and answering the research questions. This will allow you to understand where you can benefit from using live markers and test if the planned analysis approach actually works for your data. The purpose of the pilot

test is to optimize your set up, data collection and analysis protocols both when it comes to quality and time investment. During this trial it is also recommended to get a sense of how long the battery lasts. This will give you an idea of how often you will need to charge the recording unit.

Participants

When recruiting participants you want to ensure that the participants eyes are healthy and unobstructed for the eye tracking glasses recording. Generally a good guideline is to recruit participants that have

- Not undergone eye surgery
- Do not use bifocal lenses
- That do not wear contacts that block infrared light
- That do not wear strong eye makeup

Each of these factors can negatively affect the eye tracking data. Participants that do not comply with these guidelines often have issues with the calibration, and if you manage to calibrate their eyes the data quality is very likely to be negatively impacted. In short, do your best to recruit participants that have relatively healthy eyes with no history of surgery.

Pre-instructions to participants

Explain to the participant what is going to happen in practice: they will wear a set of special glasses that will measure eye movements. Explain that the glasses are wired to a small computer that records the data from

the eyes. And let them know for how long the recording session will be. At this point you may also mention that you will first go through a process to fit the frame to the participants face and it may take a bit of back and forth to get right. Before beginning, you should provide the participants with a consent form, that explains their rights with regards to being in the study. In the consent form, your participants need to be explicitly informed about what personal data is being processed by whom for what purpose as well as that they have the right to withdraw their consent at any time. Assuming the participant signs the form, the experiment can begin.

Fitting the glasses

You may want to allow the participants to sit down during the setup phase to allow them to relax more. At this point you will attach the glasses to the face of the participant. Beforehand you can assess if a different nose piece is required in order to ensure the eye-camera view of the eye is ideal. When you have used the system for a while you will be able to estimate quickly what nose piece is required by taking a quick look at the participant's face.

Depending on the type of hardware there may be a string that goes between the temple tips, that allows for tightening the fit to the head.

You may also fasten the cable that goes between the glasses frame and recording unit. If it is not fastened it is easy to accidentally pull the cable and displace the glasses.

If you plan to use prescription lenses, you will have to make sure you have already asked the participant about their prescription beforehand so you can have the right set of lenses inserted from the start.

In most cases, fitting the eye tracking glasses is as straightforward and quick as putting on any normal pair of glasses.

Calibration

At this point, you will start the calibration phase. The type of calibration you have to do will depend entirely on the hardware. Some eye tracking glasses require multi-point calibration, others a one point calibration, and yet others do not need any calibration at all. In general, if you want the highest accuracy, it is recommended to perform a manual calibration - where applicable - and visually inspect the results. For devices that do not need a calibration, performing a validation check, and a gaze correction if needed is still highly recommended.

In order to evaluate the calibration

most systems offer feedback that will allow you to make a decision to go forward or try a new calibration.

A practical tip is to use the live view from the recording unit and ask the participant to look at known targets in the environment, e.g. a clock on the wall, your finger, or something else. This will allow you to quickly assess if data looks alright. Furthermore, ask the participant to keep head still and choose targets that are across the field of view covering both center and peripheral targets - in order to gauge the accuracy and precision in the various areas of vision.

If all is well, you can hand the recording unit to the participant. Ideally the recording unit is attached e.g. to the belt or put in a pocket, so the participant has the hands free and doesn't have to worry about accidentally dropping it on the floor.

Pre Recording Instructions

Depending on the research you may want to give the participant instructions about what they are supposed to do before the recording starts. The main reason for this is



simply to minimize recording time. This will allow you to spend less time on data transfer and data analysis, and is particularly relevant if you have a large number of participants. Moreover, it will be easier to share the recordings with peers if they are smaller in size.

Executing the Recording

Now you are ready to start your recording. You have to follow the steps required for the hardware and software to get the recording started. During the recording, you can sit back and monitor the data collection live (if possible) and, if using iMotions or a similar software, use live markers to register the occurrence of events of interest. It is recommended to keep an eye on the glasses, specifically whether they start sliding. Some hardware is more notorious for sliding during recordings, and not all hardware is equally as good in correcting for changes in the eye tracking glasses position without compromising the eye tracking data quality.

Post Recording

Once the session is finished, debrief the participant. Detach cable, glasses, recording unit and so on. At this point it is strongly recommended to quickly review the recording to ensure the data has been captured properly. If the recording is long and time is sparse, then you can increase the playback speed (in iMotions you can increase it to 2x the original speed) or move the cursor to specific events of interest by dragging the timeline cursor to points of interest.

You may also want to check the battery status at this point and potentially recharge before the next participant.

Additional factors to consider during the data collection

Lighting

Try to avoid direct sunlight interfering with the recording. If strong sunlight is going through windows, try to blind them to avoid data loss.

The scene camera in most hardware platforms will have automatic gain control. It will adjust the brightness and contrast levels according to the amount of light coming through the lens. This means that the glasses will attempt to optimize the gain control depending on the lighting of the entire scene. If you suddenly have an increase in brightness, it may darken your objects of interest so much that you cannot clearly see them in the eye tracking recording replay (and thus not be able to carry out proper analysis). Some eye tracking glasses enable you to turn off the automatic gain control, and this is worth considering if you know brightness will vary a lot during your recordings.

On that note, lighting becomes particularly important to consider if you choose to leverage gaze mapping technology for your analysis. Gaze

mapping relies exclusively on contrast, and if your object of interest no longer appears in the same brightness, it will negatively affect how well the gaze mapping algorithm can identify it. This is important to test during your pilot study.

Privacy

Recording contains video and sound. You will have to ensure you only capture what is needed and receive explicit consent from the participant. For example, if your task involves the participant withdrawing money from an ATM you will potentially capture the pin code of the participant's credit card. Or if the participant is asked to perform a task on his own computer, you will capture potentially private information in that process. If you are doing an in-store study, you will capture other people in the

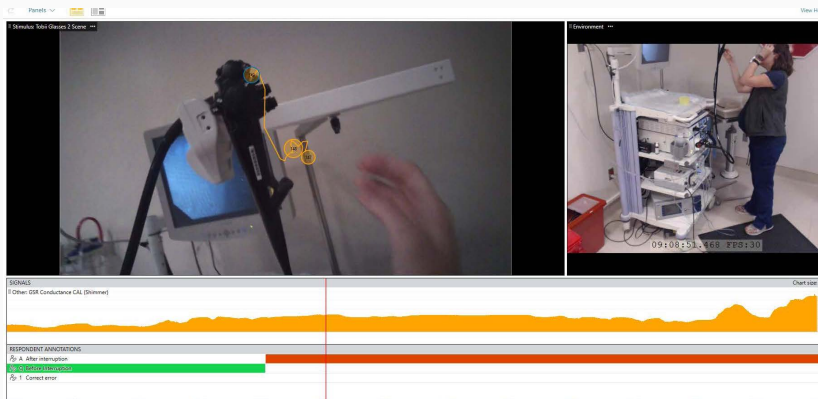
environment, and you may need explicit permission from the store to record and be mindful that other people may not want to be recorded.



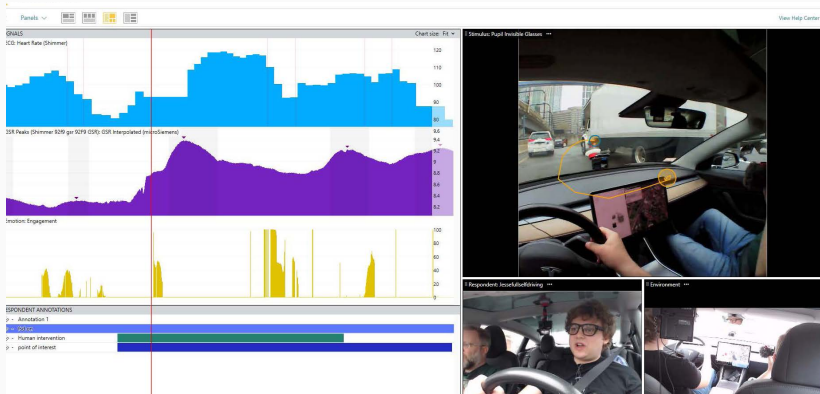
Multimodal studies

Examples of studies that synchronize eye tracking glasses with other sources of data

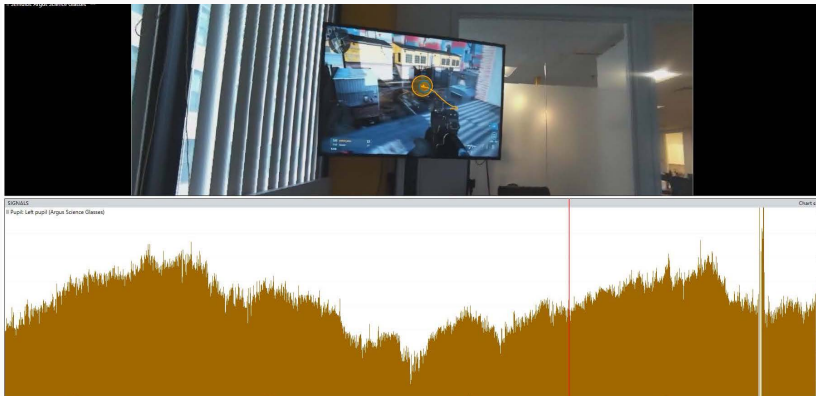
Combining eye tracking glasses data with other sensor modalities, such as heart rate, skin response, and brain activity, allows researchers to gain a comprehensive understanding of human behavior. Multimodal data integration can reveal hidden patterns and correlations, enhance insights, and increase the reliability of research findings. Physiological measures, when combined with eye tracking data, can provide valuable information about emotional and cognitive states, helping researchers contextualize gaze patterns and better understand the underlying processes driving these behaviors.



This study is about understanding how staff operate medical equipment in an OR room. It combines the first person view of the participant, external video feed to see the respondent and equipment, and a galvanic skin response trace to understand if certain tasks induce stress. As well the timeline is marked up showing when tasks begin and end.



This study aimed to understand how humans respond when the car drives autonomously. There is a view of the participants face, cabin and point of view from the glasses. Furthermore to quantify stress levels the eye tracking is combined with heart rate and skin conductance measures.



A study about video gaming, where the eye tracking glasses allow for recording the experience in a natural setting using a big screen TV, gaming console. This also allows one to understand what elements in the environment attract attention, e.g. if the player looks at the controller when challenged by the workflow of the game.

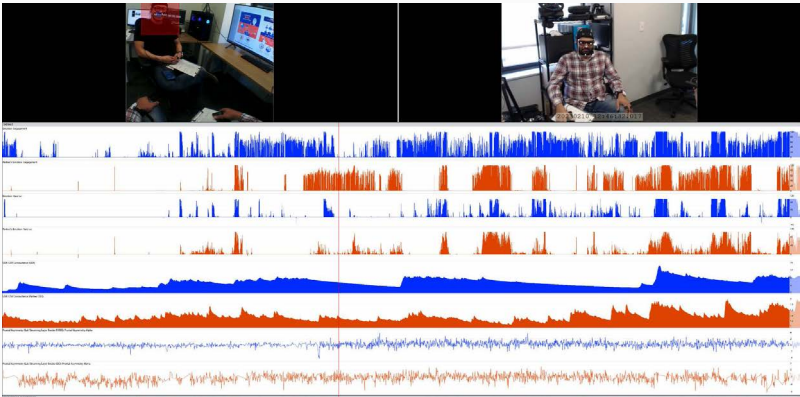
In this case the research is combining visual attention with changes in pupil size, which can be an indication of cognitive workload. A study about video gaming, where the eye tracking glasses allow for recording the experience in a natural setting using

a big screen TV, gaming console. This also allows one to understand what elements in the environment attract attention, e.g. if the player looks at the controller when challenged by the workflow of the game.

In this case the research is combining visual attention with changes in pupil size, which can be an indication of cognitive workload.



Using eye tracking glasses in urban and nature settings to understand what people pay attention to. In this case the eye tracking data is synchronized with a heart rate sensor to understand stress levels in different environments



This dyad study intended to capture the biometrics during a natural conversation. You will notice that both participants are wearing eye tracking glasses, and we are

able to correlate both of their eye tracking data to other biosensors, including GSR, EEG and FEA.

From this data set, we could quantify how the percent of time looking at a conversation partner's face correlated with intersubject neural synchrony, frontal alpha asymmetry, GSR synchrony and the percent of time they each expressed specific facial expressions, for example joy, valence and confusion. Furthermore, we were able to use voice analysis to analyze vocal features during the conversation.

Analysis of eye tracking glasses data

Congratulations, you have successfully collected all the data you need. Since you already have checked the data during the data collection to ensure the quality, there are most likely not unpleasant surprises. You now have all you need to begin analysis.

In the pilot phase you went through both data collection protocol and analysis tools, so at this point you know exactly what to do. In the following section we will present various options for analyzing eye tracking glasses data.

Qualitative review of recording

A lot can be learned from reviewing each individual recording, as much as time permits. It is very revealing to observe participants' patterns of attention, behavior, and decision making through their point of view. Especially if you are new to the wonders of eye tracking glasses, this process most certainly will give you qualitative information relevant for the present study as well as generating ideas for new ways to apply the technology in the future.

In this process you can observe not only what participants pay attention to, but also what they appear to not pay attention. The video of their point of view itself will be valuable, as you can see how they navigate the environment, when they make critical decisions related to the tasks they are given. And if you record the sound, you can hear what they answer to the moderators questions.

Task-oriented paradigms and annotations

In this analysis you may not use the eye tracking data at all. The video itself, and potentially sound, will be the foundation to annotate the timeline with events of particular interest. Typically it could be marking up on the timeline when a task starts and finished along with a comment whether the task was successful or not. For this, you can use live markers and/or annotations. Live markers are particularly useful if you have repetitive tasks that occur at different times during the recording. By marking when a task stops and ends, you can use the auto-annotation function to

easily and accurately annotate all time periods that are characterized by performing the recurrent task of interest. All of your exported data will then be annotated accordingly, and you can easily continue with your statistical analysis.

Eye tracking metrics and aggregation

Eye tracking is a powerful tool for studying visual attention. While it is relatively straightforward to analyze eye tracking data from screen-based setups, it becomes more complicated with experiments done “in the wild” using eye tracking glasses. The reason for this additional challenge is that in an eye tracking glasses recording, the visual parameters are constantly changing. In addition, if you intend to aggregate the data from multiple participants then you deal with the challenge of every participant having a unique eye tracking glasses recording. So how can you make comparisons between them?

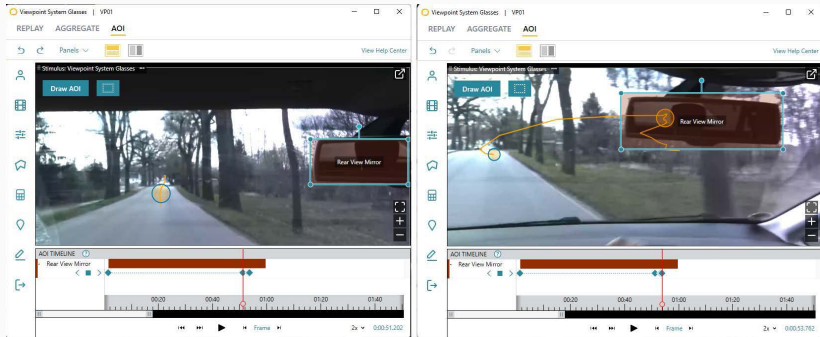
1. Moving (Dynamic) Areas of Interest (AOI)

In your study you may want to understand how many participants noticed a certain object and for how long. For example in a driving task you want to quantify how much the built-in entertainment system distracts the driver from attending the road. In this case you will review the video and place an Area of Interest on top of the entertainment system. Frame by frame you will update the position and shape of the AOI to match the position and appearance of the entertainment system console.

This can be a laborious process to do if you have many hours of recording across participants. A good AOI tool will offer ways to speed up the process, for example skipping multiple frames forward and interpolate the position of the AOI between key frames. If your dataset is very large, you can also distribute the AOI work between multiple people working in parallel.

The rear view mirror is marked with an Area of Interest, the position of the mirror in the video changes constantly and the AOI is moved by the observer to fit the position not every frame, but only when the position changes more than incrementally - the system will interpolate the position between those key moments.

You will have to repeat this process for every single participant and you will not be able to create an aggregated heatmap without using more advanced analysis tools. While using dynamic areas of interests affords you the highest precision and accuracy, they are in most cases not an efficient way to analyze eye tracking glasses data for multiple participants.

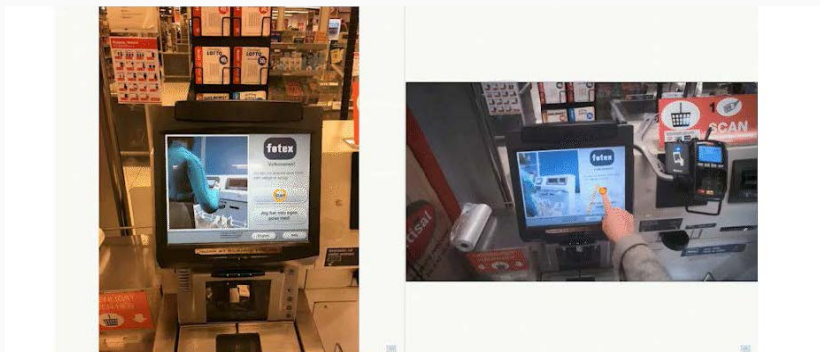


2. Gaze mapping - A Projection of dynamic data onto a static image

This problem is solved by gaze mapping. As an analytical technique offered by cutting-edge eye tracking software, it provides a way for the dynamic visual world to be represented by a static image. Let's go through exactly what this means.

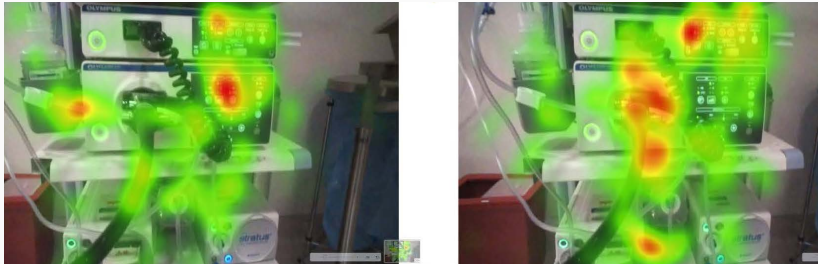
Let's say that you want to test several respondents' attention to a specific shelf in a shopping environment (although it could be any environment or setting, of course). Following instructions, each respondent could explore the environment wearing eye tracking glasses. By importing the data from all respondents into iMotions, the gaze data can be mapped onto a defined image / frame from the recording. The result is something like what is shown below.

The respondent's view is shown on the right, and the gaze is superimposed onto the image on the left.



This is completed by a reference image being analyzed and used to relate what the participant sees in the real world. The processing transposes the 3D visual scene into a 2D perspective.

It's then possible to view the same reference image with each unique participant's view – standardizing what is otherwise a different process for each.



Example of using gaze mapping on an eye tracking glasses study about training of staff in an operating room. Left image is visual attention of expert users of the equipment, while right image shows non-expert visual attention (we may need permission to use this from the client)

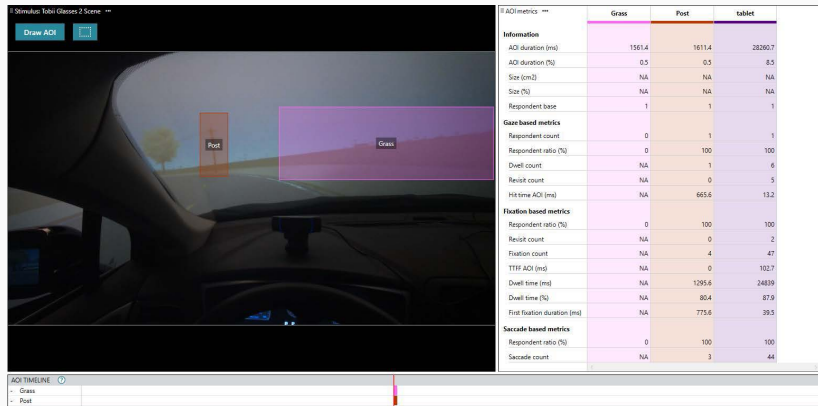
Thus, in contrast to moving AOIs, gaze mapping allows you to use static AOIs and create aggregated heatmaps. Not only do you get the ability to aggregate your eye tracking glasses data, the analysis also takes much less time.

There are instances where gaze mapping is not ideal, for example if you are tracking how people look at a moving object. In those cases, the highest precision is gained from using dynamic AOIs, although the compromise in accuracy might be circumvented by a higher sample size. These are the types of questions you should consider during your pilot experiment.

Finally, it is worth considering the overall issues with projecting a 3D video from eye tracking glasses onto a 2D image. The field of view of a scene camera on the glasses is much larger than the 2D reference image onto which the eye data is plotted. The gaze data is plotted in relation to features detected on the reference image, and may therefore look spatially closer than they were in the “real” environment. This should be kept in mind before interpreting the metrics from a gaze-mapped image, since it can negatively impact the accuracy of the output metrics. .

Key eye tracking metrics

Above we talked about the process of eye tracking analysis. What happens after? At this point you are ready to dig into the metric outputs!



A good software for data analysis should allow you to choose peer-reviewed fixation filters to ensure that your raw eye gaze data is converted accurately into fixations (and saccades), so that you can be sure that you're looking at what your participants were looking at. Typically, this is done through a **velocity-threshold identification (IV-T) filter** or a **Duration Dispersion filter**.

The IV-T filter, considered gold-standard in fixation classification, determines whether an eye is fixating on an object by analyzing the speed of eye movement compared to a velocity threshold - if the speed of the movement of the eye stays below the set threshold then it's determined to be part of a fixation, and if it moves faster then it's calculated as a saccade.

The Duration Dispersion Filter, a good alternative for trackers with lower sampling rates, determines whether the eye gaze constitutes a fixation based on how close neighboring eye gaze data points are to each other and for how long they reside in this position. Recent trends have also seen a machine learning based approach to fixation classifications, Hidden Markov Model based classification being a prominent one. A good software should provide not only the gold standards in classification, but also the flexibility to change this if required, based on different hardware and research needs. In some cases, data smoothing and cleaning may be required as well. Here, again, a good software is one that allows researchers to make informed

decisions about their smoothing and filtering process rather than a black box researchers are forced to trust with limited or no information.

After choosing the appropriate filter, you can now analyze the actual numbers. In iMotions, these can be visualized in the AOI interface, which gives a quick and easy overview of the eye tracking metrics of your interest. Here we will mention the eye tracking metrics most regularly used for the research purposes described above using the fixation-based metrics, but there are many more metrics available:

- Respondent count and respondent ratio: this metric signifies how many respondents fixated on a specific AOI. Note that this will always be 1 for dynamic AOIs that are unique to each participant. In contrast, with gaze mapping you will see the total number of respondents that fixated on a given AOI. You can compare this number to the respondent base which reports the full number of participants in the analyzed study.
- Revisit count: this metric signifies how often people keep looking back at an AOI. This is useful in many cases, for example in the automotive research example of assessing distraction by entertainment interfaces. Revisits can be interpreted as a metric of distraction and/or saliency.
- Time to first fixation (TTFF): TTFF tells you how long it took before a participant noticed an AOI. This gives you an idea of saliency, and gives you an opportunity to compare, for example, how fast participants notice an approaching car to a notification on the entertainment interface.
- Dwell time (ms and %): Dwell time is a classic eye tracking metric and simply tells you for how long someone spent looking at a particular AOI. This is useful to get an overall idea of how long participants, for example, spent looking at the road vs the entertainment interface.

It is important to note that the metrics can be calculated from the raw gaze from fixations. You can get access to both fixation and gaze-based metrics in iMotions. Further, you can also get a number of saccade and AOI based metrics allowing you to assess and control for a number of parameters related to your research question.

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